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Balanced ternary reversible comparator for qutrit quantum circuits

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Abstract

In recent years, quantum computing has emerged as a leading approach for accelerating complex computations, leveraging quantum parallelism and entanglement to surpass classical limits. Although numerous quantum circuits based on reversible gates have been proposed, optimizing circuit efficiency remains a critical challenge, particularly in the noisy intermediate-scale quantum era, where error rates and coherence times constrain performance. Ternary quantum computing, which operates on three-level quantum systems (qutrits), offers a promising alternative by reducing circuit complexity and resource overhead. In particular, balanced ternary logic presents advantages over conventional ternary logic by enhancing resource efficiency and potentially mitigating quantum gate error rates. This paper focuses on two key objectives: first, the synthesis of a reversible 1-qutrit comparator circuit using balanced ternary logic, and second, the extension to a generalized n -qutrit comparator. The proposed 1-qutrit comparator achieves notable improvements over existing unbalanced designs, demonstrating a 65% reduction in quantum cost, 50% fewer constant inputs, and a 33% decrease in garbage outputs. These results highlight the potential of balanced ternary logic to enhance the efficiency of quantum circuits, making it a valuable contribution to the development of scalable and error-resilient quantum hardware.

Keywords: quantum computing, qutrit, reversible computation, balanced ternary logic, comparator circuit

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1. Introduction

Recently, the minimization of power dissipation has become increasingly important in the design of integrated circuits, especially for digital systems. Integrated circuits that use traditional digital technology suffer from significant energy dissipation due to information loss. In 1960, Landauer demonstrated that irreversible logic involves the dissipation of $kT \ln 2$ Joules of thermal energy per bit of information lost, where k is Boltzmann's constant and T is absolute temperature [1]. Bennett has shown that the adoption of reversible logic results in a reduction in power consumption as well as energy loss during the computational process [2]. Reversible logic has gained significant attention in recent years. Using reversible gates on chips can potentially reduce energy loss [3]. Decreasing energy consumption supports worldwide environmental objectives and encourages sustainability.

Reversible logic gates are characterized as n -input n -output gates with one-to-one mapping [4–6]. In other words, a reversible gate has the same number of inputs and outputs, and vitally, it is possible to reconstruct the input vector from the output vector exactly. The main difference between reversible and classical circuits is that reversible circuits do not have feedback and fan-out [4, 7]. Low-power CMOS designs, quantum computing, and nanotechnology-based systems rely on reversible logic for their operations [4, 7, 8]. Notably, research has explored the role of reversible logic in arithmetic operations, demonstrating its impact on computational efficiency and circuit design [9].

Quantum gates are inherently reversible, as they are modelled using unitary transformations that preserve information, unlike classical irreversible gates that cause energy dissipation [7, 10, 11]. Recent advances in quantum circuit design, such as the development of techniques for decoding binary linear codes [12], categorical quantum circuits [13], and the implementation of arbitrary reversible logic gates [14], demonstrate the feasibility of constructing highly efficient quantum systems. The application of universal quantum gate sets has been demonstrated at the atomic level [15], in superconductors [16, 17], in linear optics [18, 19], and with various physical structures [20].

The hardware cost of any digital system is directly affected by the radix of number representation [21–23]. It is believed that implementing quantum computing structures based on multi-valued logic provides numerous advantages compared to their binary counterparts. Multi-valued logic reduces the number of qudits (quantum digit) more efficiently than binary logic. In multi-valued systems, ternary logic has exceptional noise immunity [24] and is also the most economical radix, which is determined by examining the width and depth of the number representation [25, 26].

Building on the advantages of ternary logic, qutrit-based quantum computation provides further benefits by leveraging a larger Hilbert space, allowing for more compact circuit representations and reducing the depth of quantum algorithms [27]. For instance, Grover's algorithm and the quantum Fourier transform can be more efficiently realized in qutrit systems, with reduced gate counts and circuit depth, potentially accelerating computation [28]. Furthermore, qutrit quantum error correction codes, such as ternary stabilizer codes, offer alternative encoding schemes that mitigate errors more efficiently than qubit-based codes [29].

Hurst [25] demonstrated that the circuit complexity (C) associated with the processing of specific quantitative data (N) can be expressed as the following equation:

$$C = K(R \cdot d). \quad (1)$$

Table 1. The relationship between values of R and C .

R	$C = K(R \cdot d)$
2	100.0
3	94.6
4	100.0
5	107.7
10	150.5

In this equation R and d are the number of basis states and the number of signals, respectively, where K is the proportional coefficient and $d = \log_R N$. It can be concluded that C reaches its lowest point when R is approximately equal to $e \approx 2.718$. Considering that R must be an integer, 3 is the nearest integer to e . Table 1 provides the relationship between the number of basis states R and their associated complexity C of digital signals.

As a comparison standard, the complexity of binary circuits is set to 100% in this table. When R is equal to 2, the value of K can be calculated using the equation below:

$$K = \frac{50}{\log_2 N} \quad (2)$$

when R is unequal to 2, the equation for C is equal to:

$$C = 50R \frac{50}{\log_R 2} = 50R \frac{\ln(2)}{\ln(R)}. \quad (3)$$

Therefore, it is reasonable to consider $R = 3$ the optimal choice, because this may result in the simplest circuit structure [30].

In addition, recent studies have found that quantum computers based on ternary logic are 37% more compact than their binary counterparts [31]. Other advantages of quantum ternary logic include increased security [32, 33], improved quantum information processing efficiency [34], greater density of stored information [35], improved error tolerance [36], simplified inter-connection complexity, and reduced power consumption [35, 37].

Qutrit-based implementations of the Deutsch–Jozsa and Grover’s algorithms also require fewer quantum gates and shallower circuit depth compared to qubit-based equivalents, accelerating execution time [38]. In cryptographic applications, qutrit gates allow for more efficient information encoding, reducing the number of required operations [39]. Preliminary research suggests that ternary stabilizer codes can offer more efficient syndrome extraction and fault-tolerant procedures than binary stabilizer codes [29].

However, implementing qutrit-based quantum computation comes with increased hardware complexity. The control and readout of qutrit states require precise calibration, as leakage into higher energy levels introduces additional noise. Superconducting circuits, such as transmon qubits, can theoretically support qutrits, but the $|2\rangle$ state exhibits shorter coherence times and greater susceptibility to decoherence [40]. Moreover, existing quantum processors and quantum error correction schemes are primarily designed for qubit-based architectures, meaning that a shift toward qutrit-based computing would require extensive redesign of fault-tolerant protocols and quantum compilers [27]. Although qubits offer promising computational advantages, their practical implementation remains a significant challenge. One of the most critical issues is the presence of noisy outputs, which are exacerbated by the increased susceptibility of higher-dimensional quantum states to decoherence. Consequently, errors in

qutrit circuits can be more complex to detect and correct than those in qubit-based systems. To mitigate these problems, researchers have proposed several strategies, including qutrit-specific quantum error correction codes, optimized pulse-shaping techniques, hybrid qubit–qutrit encoding schemes, and dynamical decoupling methods to improve coherence times [41, 42]. These approaches aim to reduce the impact of noise and stabilize qutrit operations in real-world environments. Although balanced ternary logic can reduce circuit depth and resource usage, its robustness under realistic noise conditions remains an open research question [24]. Future efforts should focus on developing adaptive error mitigation techniques and circuit-level optimizations to fully harness the potential of qutrit-based quantum computing.

In quantum ternary systems, each unit of information is known as a qutrit. Reversible ternary logic can theoretically be used to design quantum computing structures [43]. Thus, researchers have recently focused a great deal of attention on reversible ternary logic, resulting in numerous contributions in the literature concerning ternary reversible circuit designs [44–47].

Ternary arithmetic functions can be represented in two main ways: unbalanced and balanced. In the unbalanced representation, a qutrit can be depicted by $|0\rangle$, $|1\rangle$, or $|2\rangle$ values, whereas in the balanced representation, a qutrit can be represented by values of $|\hat{1}\rangle$, $|0\rangle$, or $|1\rangle$. Balanced ternary representation offers mathematical symmetry that simplifies arithmetic operations and circuit structures.

Balanced ternary representation provides notable advantages over the unbalanced form in arithmetic circuit design. It allows for simpler ternary inversion, eliminates the need for an extra sign digit, and enables efficient rounding and truncation. Balanced representation also simplifies the generation of partial products and reduces the probability of carry propagation, resulting in shorter carry ripple structures [24]. These properties make balanced ternary logic an attractive choice for implementing efficient arithmetic functions in quantum ternary circuits. Motivated by the efficiency and compactness offered by balanced ternary logic, we propose a novel circuit design leveraging this representation.

Quantum comparators are fundamental components in quantum computing, enabling efficient arithmetic operations, error correction, search algorithms, and decision making in quantum machine learning. In ternary quantum systems, they offer significant advantages over binary comparators by requiring fewer qudits and gates, which reduces circuit depth and improves resource utilization. This efficiency is essential for scalable and fault-tolerant quantum architectures. To the best of our knowledge, all ternary quantum reversible comparator circuits have been previously introduced using the unbalanced representation [48–52]. In this work, we introduce a novel balanced ternary reversible 1-qutrit comparator circuit, utilizing the balanced representation. The design is further extended to an n -qutrit comparator, achieving improvement in quantum cost, constant inputs, and garbage outputs compared to previous unbalanced designs. These metrics are defined as follows:

- *Quantum cost* refers to the number of ternary primitive gates, which are 1-qutrit Shift gates and 2-qutrit Muthukrishnan-Stroud gates, required to realize the circuit.
- *The number of garbage outputs* indicates the outputs which are generated to preserve one-to-one mappings but have unimportant values.
- *The number of constant inputs* signifies the number of inputs which must be maintained constant in order to synthesize the specified logic function.

Minimization of the above mentioned features increases the efficiency of ternary quantum reversible logic design [53, 54]. In this work, we aim to minimize these factors as key performance criteria for optimizing ternary quantum reversible comparator circuits. Our design achieves a 65% reduction in quantum cost, 50% fewer constant inputs, and 33% fewer garbage

outputs compared to previous unbalanced designs, demonstrating significant improvements in resource efficiency.

The structure of this paper is as follows: the preliminary concepts of balanced ternary logic, balanced ternary Galois field logic, along with the balanced ternary primitive gates are outlined in section 2. Our proposed realization of balanced ternary comparator circuits is presented in section 3. We discuss the evaluation of the proposed circuits and present the comparison results in section 4. Finally, the conclusion of this work is given in section 5.

2. Preliminaries

In this section, we provide the foundational concept on balanced ternary logic, balanced ternary Galois fields, and balanced ternary gates which will be used in the following sections.

2.1. Background of balanced ternary logic

In quantum ternary systems, each information unit called qutrit can be describe by three primary states. The balanced ternary representation considers $|\hat{1}\rangle$, $|0\rangle$ and $|1\rangle$ as the three primary states of a qutrit [24]. The states can be observed in equation (4), which are represented by three 3×1 matrix, and their graphical representation in a three-dimensional state space is shown in figure 1.

$$|\hat{1}\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, |0\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (4)$$

In the balanced ternary representation, a qutrit can exist in a superposition of three states simultaneously through a linear combination of the computational basis states $|\hat{1}\rangle$, $|0\rangle$, and $|1\rangle$. This superposition denoted by ψ , which is defined as $\psi = \alpha|\hat{1}\rangle + \beta|0\rangle + \gamma|1\rangle$.

The probabilities of occurrence for each of these states are $|\alpha|^2$, $|\beta|^2$, and $|\gamma|^2$, respectively, and they satisfy the condition $|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1$.

An n -qutrit system shows 3^N unique computational basis states, which can be illustrated as $|\hat{1}\hat{1}\dots\hat{1}\rangle$, $|\hat{1}\hat{1}\dots 0\rangle$, ..., $|11\dots 1\rangle$. The tensor product of n -qutrits can be used to represent all possible states within the n -qutrit system. For example, the states $|\hat{1}\hat{1}\rangle$, $|\hat{1}0\rangle$, $|\hat{1}1\rangle$, $|0\hat{1}\rangle$, $|00\rangle$, $|01\rangle$, $|1\hat{1}\rangle$, $|10\rangle$ and $|11\rangle$ represent a balanced 2-qutrit system.

2.2. Background of balanced ternary Galois field logic

In balanced ternary logic, the algebraic structure of Galois field (GF3) is defined by the set of elements $T = \{\hat{1}, 0, 1\}$, along with the two operations: addition ($\oplus$) and multiplication ($\odot$) [55]. These operations are illustrated in tables 2 and 3.

The following axioms rules are also derived according to the tables provided above, which demonstrate the behaviour of addition and multiplication operations in the algebraic structure [55].

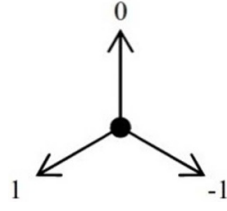
Addition:

(A1) Associative law: $a \oplus (b \oplus c) = (a \oplus b) \oplus c$

(A2) Commutative law: $a \oplus b = b \oplus a$

(A3) Identity element: There is an element 0 such that $a \oplus 0 = a$ for all a

(A4) Additive inverse: For any a , there is an element $(-a)$ such that $a \oplus (-a) = 0$

**Figure 1.** Balanced ternary representation.**Table 2.** Truth table of GF3 addition operation.

$\oplus$	$\hat{1}$	0	1
$\hat{1}$	1	$\hat{1}$	0
0	$\hat{1}$	0	1
1	0	1	$\hat{1}$

Table 3. Truth table of GF3 multiplication operation.

$\odot$	$\hat{1}$	0	1
$\hat{1}$	1	0	$\hat{1}$
0	0	0	0
1	$\hat{1}$	0	1

Multiplication:

(M1) Associative law: $a \odot (b \odot c) = (a \odot b) \odot c$

(M2) Commutative law: $a \odot b = b \odot a$

(M3) Identity element: There is an element 1 (not equal to 0) such that $a \odot 1 = a$ for all a

(M4) Multiplicative inverse: For any $a \neq 0$, there is an element a^{-1} such that $a \odot a^{-1} = 1$

Distribution:

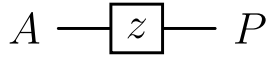
(D) Distributive law: $a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$

2.3. Balanced ternary Shift gates

In balanced ternary reversible logic, any transformation of the qutrit is expressed using a 3×3 unitary matrix, as demonstrated in figure 2. Each column or row in these matrices contains three elements which are 0 or 1. The transformations are used to shift or exchange elements within the matrices [56, 57]. $Z(+0)$ is an initial state and each column or row of the matrix shows $|\hat{1}\rangle$, $|0\rangle$ or $|1\rangle$ values in equation (4). $Z(+1)$ shifts the qutrit state $|\hat{1}\rangle$ and $Z(+2)$ shifts the qutrit state $|1\rangle$. $Z(0\hat{1})$ exchanges the qutrit states $|\hat{1}\rangle$ and $|0\rangle$, $Z(\hat{1}0)$ exchanges the qutrit states $|\hat{1}\rangle$ and $|0\rangle$, and $Z(\hat{1}1)$ exchanges the qutrit states $|\hat{1}\rangle$ and $|1\rangle$.

Each of these unitary matrices can be realized as a 1-qutrit Shift gate. These gates are uncontrollable, this means that there is no controlling input. The symbolic representation of ternary shift gates can be observed in figure 3. As depicted, A is the input to the gate and P is the output. P corresponds to the Z -transform of A . The quantum cost associated with these gates is 1 [20]. Table 4 shows the truth table of balanced ternary Shift gates, named Identity, Single Shift, Dual Shift, Self Shift, Self Single Shift and Self Dual Shift, respectively. As shown in

$$\begin{aligned}
Z(0) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & Z(+1) &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} & Z(+2) &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \\
Z(01) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} & Z(\hat{1}0) &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} & Z(\hat{1}\hat{1}) &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}
\end{aligned}$$

Figure 2. Balanced representation of 1-qutrit permutative transforms.**Figure 3.** The symbolic representation of ternary shift gates.**Table 4.** Truth table of balanced ternary Shift gates.

Input	Z(0)	Z(+1)	Z(+2)	Z(01)	Z($\hat{1}0$)	Z($\hat{1}\hat{1}$)
$\hat{1}$	$\hat{1}$	0	1	$\hat{1}$	0	1
0	0	1	$\hat{1}$	1	$\hat{1}$	0
1	1	$\hat{1}$	0	0	1	$\hat{1}$

Table 5. Inverse gates of balanced ternary Shift gates.

Gates	Z(+1)	Z(+2)	Z(01)	Z($\hat{1}0$)	Z($\hat{1}\hat{1}$)
Inverse gates	Z(+2)	Z(+1)	Z(01)	Z($\hat{1}0$)	Z($\hat{1}\hat{1}$)

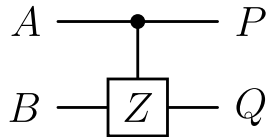
**Figure 4.** The symbolic representation of ternary Muthukrishnan and Stroud gate.

table 5, these gates have unitary inverse gates that return the inputs to their original values [20]. As can be observed in this table, $Z(01)$, $Z(\hat{1}0)$ and $Z(\hat{1}\hat{1})$ are self-inverse gates.

2.4. Balanced ternary Muthukrishnan and Stroud gate

Muthukrishnan and Stroud introduced a family of 2-qudit ternary gates that can be theoretically implemented using quantum technologies, such as liquid ion trap technology, as an elementary gate [43]. The symbolic representation of the balanced ternary Muthukrishnan and Stroud gates can be observed in figure 4. As depicted, this gate is controllable, A and B are the inputs to the gate, A is the controlling input, and B is the controlled input. P and Q are the outputs. The output P corresponds to the input A and the output Q corresponds to the Z -transform of

Table 6. Truth table of a *1-qutrit comparator* circuit.

A	B	F
$\hat{1}$	$\hat{1}$	0
$\hat{1}$	0	$\hat{1}$
$\hat{1}$	1	$\hat{1}$
0	$\hat{1}$	1
0	0	0
0	1	$\hat{1}$
1	$\hat{1}$	1
1	0	1
1	1	0

B where $Z \in \{0, +1, +2, 01, \hat{1}0, \hat{1}1\}$, but only when the input A is equal to $|1\rangle$, otherwise the output P is equal to the input B . The quantum cost associated with this gate is 1 [20].

3. Proposed design of balanced ternary comparator circuits

In this section, we present a *1-qutrit comparator* based on balanced ternary reversible logic. We use this comparator design to construct *n-qutrit balanced ternary reversible comparator*. The proposed designs in this study utilize balanced Shift gates and Muthukrishnan and Stroud gates.

3.1. Balanced ternary 1-qutrit Comparator circuit

Ternary comparators play a critical role in balanced ternary reversible networks and are an integral part of complex devices. Table 6 presents the truth table of a *1-qutrit comparator* circuit. This table shows that A and B are the input variables, and F represents the output variable. When the inputs A and B are equal, the output F is 0. When A is greater than B , the output F is 1. In contrast, if A is less than B , the output F is equal to $\hat{1}$. Generally, the function F is defined in equation (5):

$$F = \begin{cases} \hat{1} & \text{if } A < B \\ 0 & \text{if } A = B \\ 1 & \text{if } A > B. \end{cases} \quad (5)$$

In order to construct balanced ternary *1-qutrit comparator*, we used ternary balanced Self Shift gate and ternary balanced Muthukrishnan and Stroud gate. The proposed balanced ternary comparator can be observed in figure 5. This circuit takes two single qutrit numbers as inputs and assigns a value to the output based on the relationship between the input values. If the first input is equal to the second one, the output is set to $|0\rangle$. If the first input is greater than the second one, the output is set to $|1\rangle$. If the first input is less than the second input, the output is equal to $|\hat{1}\rangle$. As can be seen in figure 5, in our realization, there are three inputs A, B and $|0\rangle$. P, Q and F represent the output of the circuit. The output F , which is the target output, represents the result of the comparison of inputs A and B . This realization requires one *constant input*, which is $|0\rangle$, and produces two *garbage outputs*, which are P and Q . To design this circuit, we used 5 Muthukrishnan and Stroud gates and 4 shift gates, resulting in a *quantum cost* of 9 for the proposed realization. However, if input restoration is not needed, the last Shift gate

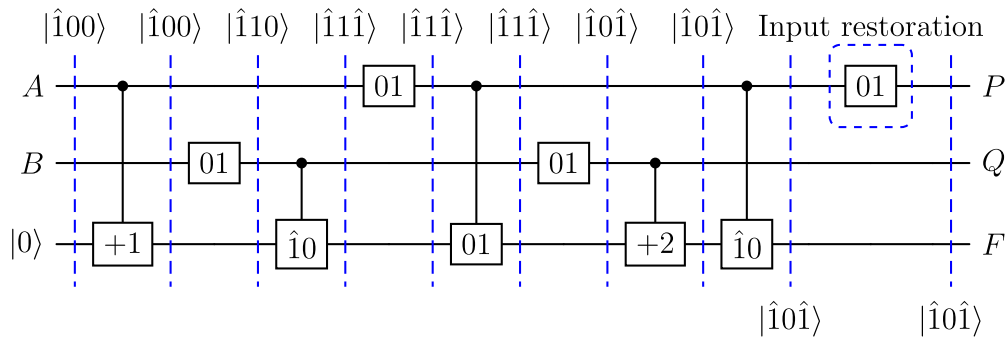


Figure 5. The proposed balanced ternary reversible *l*-qutrit comparator circuit.

can be removed, and the quantum cost decreases to 8. The following example illustrate the functionality of the proposed balanced ternary 1-qutrit comparator circuit:

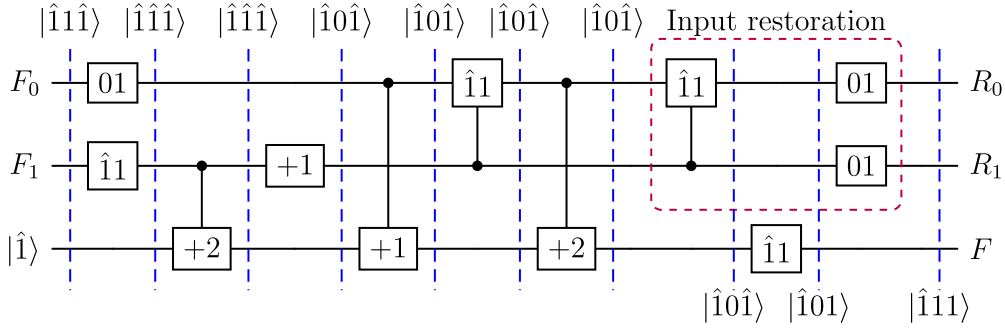
Example 1. Consider the case where the inputs are $A = |\hat{1}\rangle$ and $B = |0\rangle$. As illustrated in figure 5, input A passes through the circuit unchanged and appears at output P as $|\hat{1}\rangle$. The input B , on the other hand, is temporarily transformed to $|1\rangle$ during the second step of the circuit in the figure. However, in the sixth step, its original value is restored, and it exits the circuit at output Q as $|0\rangle$. The constant ancilla input $|0\rangle$ evolves based on the values of A and B . In the third step, since B has been set to $|1\rangle$, the ancilla changes to $|\hat{1}\rangle$. This value remains unchanged through the subsequent gates, as they do not affect it in this particular input condition. Therefore, the final output $F = |\hat{1}\rangle$ and according to equation (5), this result correctly indicates that $A < B$. This specific case corresponds to one of the entries in table 6, where the output $F = |\hat{1}\rangle$ denotes that the comparator has correctly evaluated the ternary condition $A < B$.

3.2. Balanced ternary n -qutrit comparator circuit

In a balanced ternary n -qutrit comparator, two n -qutrit numbers are being compared. At the beginning, each qutrit of these numbers is compared using n separated 1-qutrit comparators simultaneously. In the next step, the results from these 1-qutrit comparators are compared, beginning with the least significant qutrits. In the second step, we use $n - 1$ *sub comparator circuit* to compare the results of the first step.

The proposed realization for the *sub comparator circuit* can be observed in figure 6. In this circuit, three inputs F_0 , F_1 and $|\hat{1}\rangle$. F_0 and F_1 are the outputs of the previous step and $|\hat{1}\rangle$ is the constant input. R_0 , R_1 and F represent the outputs of the circuit. The outputs R_0 and R_1 can be equal to F_0 and F_1 , respectively. The output F , which is the target output, represents the result of the inputs F_0 and F_1 comparison. This realization requires one *constant input*, which is $|\hat{1}\rangle$, and produces two *garbage outputs*, which are R_0 and R_1 . The number of Shift gates and Muthukrishnan and Stroud gates is 6 and 5, respectively. Therefore, the *quantum cost* associated with the proposed realization is 11. However, if input restoration is not required, the Shift and Muthukrishnan and Stroud gates in the box can be removed, reducing the quantum cost to 8. The following example is presented to demonstrate and clarify the functional behaviour of the proposed balanced ternary sub comparator circuit. This example serves to illustrate how the circuit processes different input combinations and produces the corresponding outputs.

Example 2. Let the inputs be $F_0 = |\hat{1}\rangle$ and $F_1 = |1\rangle$. As shown in the circuit in figure 6, input F_1 is first transformed into $|\hat{1}\rangle$ in the first step. In the third step, it undergoes a $(+1)$ shift

**Figure 6.** The proposed balanced ternary reversible *sub comparator* circuit.**Table 7.** Truth table of the sub comparator circuit.

F_0	F_1	F
$\hat{1}$	$\hat{1}$	$\hat{1}$
$\hat{1}$	0	$\hat{1}$
$\hat{1}$	1	1
0	$\hat{1}$	$\hat{1}$
0	0	0
0	1	1
1	$\hat{1}$	$\hat{1}$
1	0	1
1	1	1

gate, changing its value to $|0\rangle$. Later, in the final stage, a (01) shift gate is applied, restoring the original value. As a result, the output R_1 becomes $|1\rangle$. Meanwhile, input F_0 remains unchanged throughout the computation and appears directly at output R_0 as $|\hat{1}\rangle$. The constant ancilla input, initialized as $|\hat{1}\rangle$, is modified in step eight to $|1\rangle$, and this value is passed to the target output F . This result aligns with the corresponding entry in table 7, confirming the correct operation of the ternary sub comparator circuit.

Table 7 shows the truth table of the *sub comparator circuit*. As can be seen in this table, if the first number is equal to the second, the output is set to 0, when F_0 and F_1 are equal to 0. If the first number is greater than the second one, the output is set to 1 when F_1 is 1, or F_0 and F_1 are equal to 1 and 0, respectively. If the first number is less than the second number, the output is equal to $\hat{1}$, when F_1 is $\hat{1}$, or F_0 and F_1 are equal to $\hat{1}$ and 0, respectively. The realization of our proposed *balanced ternary n-qutrit comparator* circuit is shown in figure 7. To construct this circuit, we used n 1-qutrit comparator circuits and $n - 1$ sub comparator circuits, which are denoted by *1-qutrit Comp* and *Sub Comp*, respectively. The quantum cost, the number of constant inputs and garbage outputs can be determined according to equations (6)–(8), respectively.

$$\text{Quantum cost} = 8n + 8(n - 1) = 16n - 8 \quad (6)$$

$$\text{The number of constant inputs} = n + n - 1 = 2n - 1 \quad (7)$$

$$\text{The number of garbage outputs} = 2n + 2(n - 1) = 4n - 2 \quad (8)$$

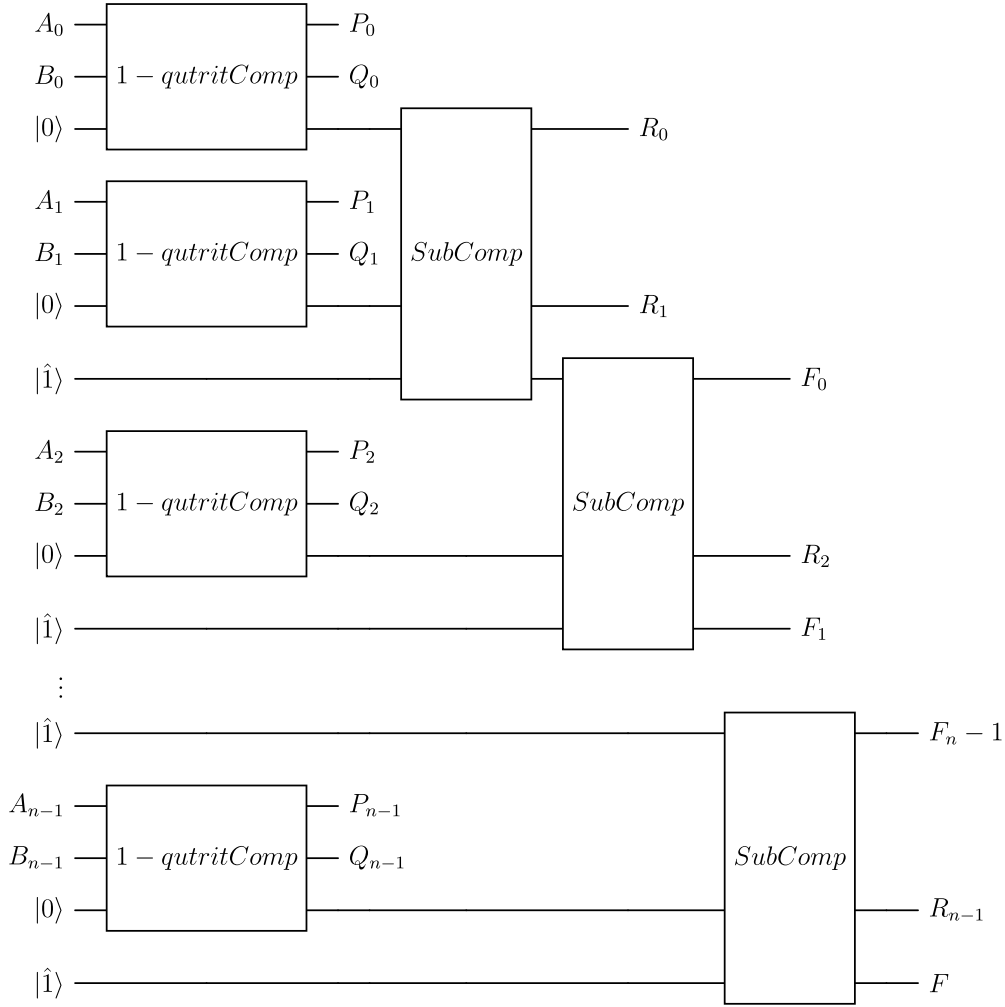


Figure 7. The proposed balanced ternary reversible n -qutrit comparator circuit.

As an example, consider two balanced ternary 2-qutrit numbers, A and B . For the comparison between A and B , in the first step, A_0 and A_1 should be compared using our proposed 1-qutrit comparator with B_0 and B_1 , respectively. In the next step the results of 1-qutrit comparator circuits should be compared using the proposed sub comparator circuit. The final result is in accordance with table 7, equations (9)–(11):

$$A_1A_0 = B_1B_0 \text{ if } A_1 = B_1 \text{ and } A_0 = B_0 \quad (9)$$

$$A_1A_0 > B_1B_0 \text{ if } A_1 > B_1 \text{ or } A_1 = B_1 \text{ and } A_0 > B_0 \quad (10)$$

$$A_1A_0 < B_1B_0 \text{ if } A_1 < B_1 \text{ or } A_1 = B_1 \text{ and } A_0 < B_0 \quad (11)$$

According to equation (9), the final result is set to $|0\rangle$ when both results in the first step are equal to $|0\rangle$. Equation (10) shows that the final result is set to $|1\rangle$, if the result of second comparator is $|1\rangle$ or the result of the first and the second are $|1\rangle$ and $|0\rangle$, respectively. Equation (11) also means that the final output is set to $|\hat{1}\rangle$, if the result of second comparator is $|\hat{1}\rangle$ or the

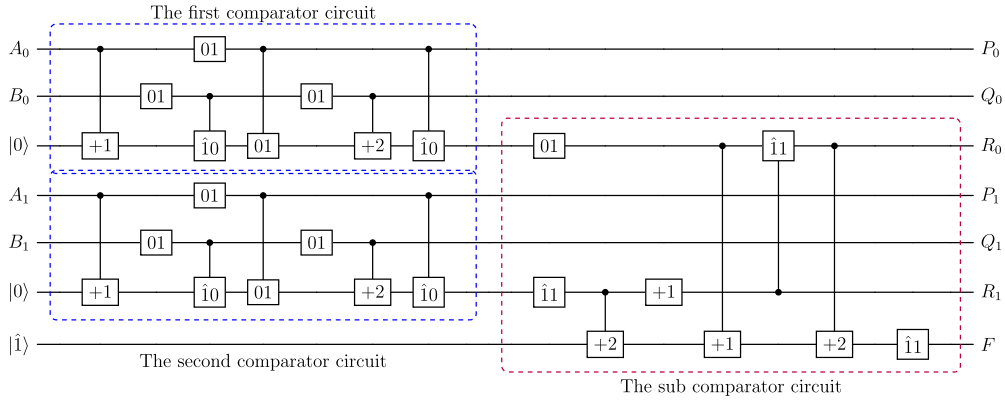


Figure 8. The proposed balanced ternary reversible 2-qutrit comparator circuit.

result of the first and the second are $|\hat{1}\rangle$ and $|0\rangle$, respectively. The realization of balanced ternary 2-qutrit comparator circuit is illustrated in figure 8. As you can see, the inputs of the first comparator circuit are A_0 and B_0 , and for the second comparator the inputs are shown by A_1 and B_1 . The target outputs of these circuits are the inputs of the sub comparator circuit. This comparison requires 3 constant inputs and produces 6 garbage outputs, which align with the equations (7) and (8), respectively. The quantum cost of this circuit is 24 which is in accordance with equation (6). To better understand the operation of the proposed balanced ternary 2-qutrit comparator circuit, the following example is provided. It highlights how the circuit behaves under different input conditions and demonstrates its ability to accurately compare ternary values.

Example 3. As a representative case to evaluate the behaviour of the proposed 2-qutrit comparator circuit in figure 8, let us consider the ternary inputs $A_1A_0 = |00\rangle$ and $B_1B_0 = |01\rangle$. The first 1-qutrit comparator circuit evaluates the least significant qutrit: $A_0 = |0\rangle$ and $B_0 = |1\rangle$. The comparison result is $|\hat{1}\rangle$, indicating that $A_0 < B_0$. The second 1-qutrit comparator circuit compares the most significant qutrit: $A_1 = |0\rangle$ and $B_1 = |0\rangle$. Since both are equal, the result is $|0\rangle$, indicating $A_1 = B_1$. These results are consistent with equation (5) and the values shown in table 6. In the second stage, the sub-comparator circuit in the red box receives the outputs of the two 1-qutrit comparators in the blue boxes, which are $|\hat{1}\rangle$ and $|0\rangle$, respectively. Based on this, the final comparison output is $F = |\hat{1}\rangle$, correctly indicating that $A < B$. This result aligns with the logic expressed in equation (11), and also matches the expected output in table 7, further validating the circuit functionality.

4. Evaluations and discussion

In this work, we introduced the use of balanced representation logic for ternary comparator circuits for the first time, leading to a substantial improvement in overall performance. This section involves two separate assessments where we compare our proposed comparator circuits with the existing unbalanced ternary counterparts to assess their relative performance and efficiency. Firstly, we examine the quantum cost of the proposed 1-qutrit comparator alongside the number of constant inputs and garbage outputs. These factors are thoroughly examined,

Table 8. Comparison between proposed 1-qutrit comparator and unbalanced designs.

	Proposed comparator	Improvement percentage	[50]	[52]	[51]
Quantum cost	8	65%	23	23	55
Constant input number	1	50%	2	2	6
Garbage output number	2	33%	3	3	7

Table 9. Comparison between proposed n -qutrit comparator and unbalanced designs.

	Proposed comparator	[48]	[52]	[49]	[50]	[51]
Quantum cost	$16n - 8$	$18n - 8$	$34n - 11$	$236n - 169$	$43n - 20$	$56n + 6$
Constant input number	$2n - 1$	$2n - 1$	$3n - 1$	$13n - 7$	$4n - 2$	$5n + 3$
Garbage output number	$4n - 2$	Not reported	Not reported	Not reported	Not reported	Not reported

analysed, and compared with their unbalanced counterparts [50–52]. Secondly, the proposed n -qutrit comparator is evaluated objectively and comprehensively with other existing ones [48–52]. It should be noted that in the first part of assessments, we determine the improvement rate concerning the best results in the literature.

4.1. Evaluation of ternary reversible 1-qutrit comparator circuits

In table 8, we provide a comparative analysis for ternary reversible 1-qutrit comparator with respect to the quantum cost, the number of constant inputs, the number of garbage outputs. In particular, this comparison demonstrates that the quantum cost of our proposed design is 65% lower than the unbalanced versions [50, 52]. The proposed circuit has reduced the number of garbage outputs from 3 to 2, representing a reduction 33%. The table also shows an improved efficiency in terms of the number of constant inputs which decreased from 2 to 1, resulting in an improvement in 50%. It can be concluded that our design is more efficient than its unbalanced counterparts in the literature [50–52].

4.2. Evaluation of ternary reversible n -qutrit comparator circuits

Table 9 provides a comprehensive comparison between our proposed balanced ternary n -qutrit comparator circuit and existing alternatives [48–52]. This comparison clearly illustrates that the proposed comparator, in balanced representation, surpasses others in [48–52]. Mainly because it has less quantum cost and fewer constant input number and garbage output number used in the design, leading to considerable improvements. It should be noted that the number of garbage output in our design is $4n - 2$ and in [49–52] it is unreported. However, the

undeniable fact is that, due to the considerably higher number of constant inputs compared to our design which has $2n - 1$ constant inputs, the number of garbage output is significantly greater. It is also important to note that, since the lower quantum cost leads to better efficiency in quantum circuit design, despite having the same number of constant inputs as the proposed design in [48] with the value of $2n - 1$, the lower quantum cost in our design, which is $16n - 8$ allows us to assert its greater efficiency.

5. Conclusion

In this work, we proposed a balanced ternary reversible 1-qutrit comparator circuit using Shift gates and the Muthukrishnan-Stroud gate. To support scalability, we introduced a balanced ternary reversible sub-comparator, leading to a generalized n -qutrit comparator. Our results show significant improvements in quantum cost, constant inputs, and garbage outputs compared to unbalanced designs, highlighting the potential of balanced ternary logic in qutrit-based quantum computing. Although ternary quantum computing offers efficiency gains, its implementation faces challenges. The instability of the $|2\rangle$ state in superconducting qubits increases decoherence, making qutrit operations more error-prone. Additionally, current hardware lacks native support for high-fidelity qutrit gates, requiring advances in control techniques and noise mitigation. Despite these challenges, qutrit-based algorithms, such as Grover's search and the quantum Fourier transform, show promise in reducing circuit depth. Further research is needed to determine when qutrit computing provides a practical advantage and to develop qutrit-specific error correction frameworks. The improvements presented in this work are at the logical circuit level; future experimental validation is essential to assess performance under realistic noise conditions. The proposed circuits offer a foundation for future advancements in balanced ternary quantum systems, paving the way for more efficient quantum algorithms, improved error resilience, and novel applications in quantum computing.

Data availability statement

All relevant data are included in the article.

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